PART 1 : PHYSICS

SACTION-A

1) Consider the potential at the corner of a uniformly charged cube of dimension L to be directly proportional to ρL^2, where ρ is volume charge density. What is the value of the ratio of the potential at the centre to the potential at the corner of the cube :-

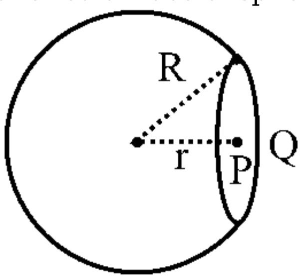
(A) 4 : 1

(B) 3 : 2

(C) 1 : 2

(D) 2 : 1

2) Point charge Q is placed at point P in the plane of circular face of sphere. The electric flux passing through the spherical part shown in the figure is :-



(A) $\frac{Q}{\epsilon_0}$

(B) $\frac{Q}{2\epsilon_0}$

(C) $\frac{Q}{3\epsilon_0}$

(D) Can not be calculated.

3) Two point charge of 100 μC and -4 μC are positioned at points (-2√3, 3√3, -4) and (4√3, -5√3, 6) respectively of a Cartesian coordinate system. Find the force vector on the -4 μC charge? All the coordinates are in meters.

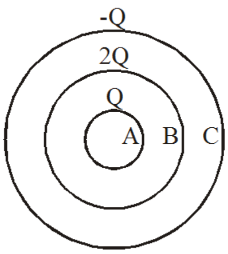
(A) $9 \times 10^{-4}\left(3\sqrt{3}\hat{i} - 4\sqrt{3}\hat{j} + 5\hat{k}\right)$

(B) $9 \times 10^{-4}\left(-3\sqrt{3}\hat{i} + 4\sqrt{3}\hat{j} - 5\hat{k}\right)$

(C) $2.25 \times 10^{-4}\left(-3\sqrt{3}\hat{i} + 4\sqrt{3}\hat{j} - 5\hat{k}\right)$

(D) $2.25 \times 10^{-4}\left(3\sqrt{3}\hat{i} - 4\sqrt{3}\hat{j} + 5\hat{k}\right)$

4) Charge Q, 2Q and –Q are given to three concentric conducting spherical shells A, B and C respectively. The ratio of charges on the inner and the outer surfaces of the shell ‘C’ will be :-



(A) $\frac{3}{4}$

(B) $-\frac{3}{4}$

(C) $\frac{3}{2}$

(D) $-\frac{3}{2}$

5) Mohit says that Gauss's Law can be used to find the electric field of a sufficiently symmetrical distribution of charge as long as $\vec{E} \cdot d\vec{A}$ over the whole Gaussian surface. Anurag says that the electric field must be a constant vector over the entire Gaussian surface. Which one, if either, is correct?

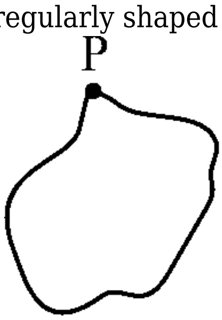
(A) Mohit, because that means no charge is enclosed within the Gaussian surface.

(B) Anurag, because a constant electric field means that $\oint \vec{E} \cdot d\vec{A} = 0$.

(C) Both, because the conditions in (1) and (2) are equivalent.

(D) Neither, because the electric field can be found from Gauss's law even if $\vec{E} \cdot d\vec{A} \neq 0$ over the whole Gaussian surface.

6) If we have an irregularly shaped conductor as shown and a charge is given to it, choose the correct statement.



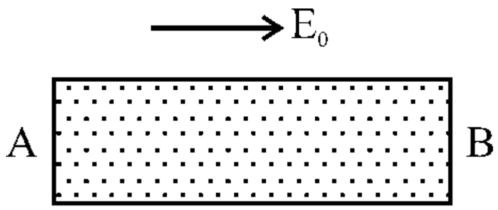
(A) the electric field and electric potential at P is maximum among all points of conductor.

(B) the electric field at P is maximum and electric potential is same at all points of conductor.

(C) the electric field and electric potential at P is same at all points of conductor.

(D) the electric field and electric potential at P is minimum among all points of conductor.

7) A metallic rod is placed in a uniform electric field. Select the **CORRECT** option.



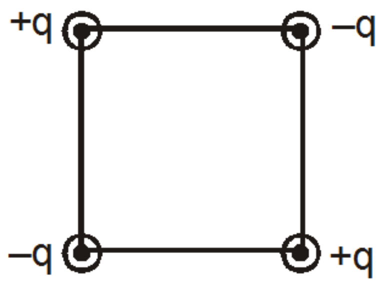
(A) Inside the rod there will be an induced electric field from B to A which cancels external uniform field in the rod.

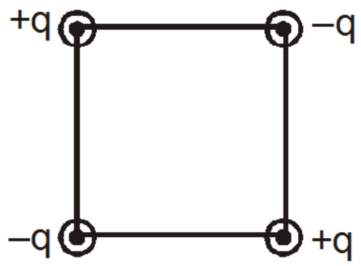
(B) Free electrons will accumulate at the end B of the rod.

(C) The potential of the end A will be more than that at B.

(D) The electric field outside the rod will not change due to the induced charges in the rod.

8) **Statement-1** : The electric potential and the electric field intensity at the centre of a square having four point charges at their vertices (as shown) are zero.





Statement-2 : Electric field is negative derivative of the potential.

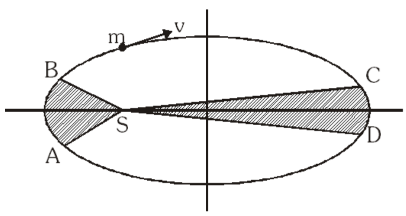
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

9) The figure shows elliptical orbit of a planet m about the sun S. The shaded area SCD is twice the shaded area SAB. If t_1 be the time for the planet to move from C to D and t_2 is the time to move from A to B, then :



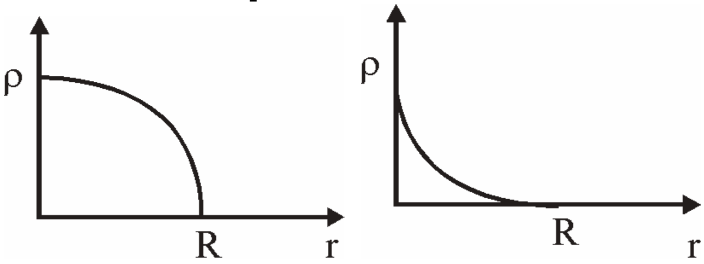
(A) $t_1 = t_2$

(B) $t_1 = 8t_2$

(C) $t_1 = 4t_2$

(D) $t_1 = 2t_2$

10) Two planets A and B have same mass and radii(R). The variation of density of the planets with distance from centre is shown in the following diagrams. The ratio of acceleration due to gravity at the surface of the planets A and B will be :



(A) > 1

(B) < 1

(C) 1

(D) will depend on the maximum density

11) Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : The escape velocities of planet A and B are same. But A and B are of unequal mass.
Reason R : The product of their mass and radius must be same, $M_1R_1 = M_2R_2$
In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are correct but R is not the correct explanation of A.

(B) A is correct but R is not correct.

(C) Both A and R are correct and R is the correct explanation of A.

(D) A is not correct but R is correct.

12) A shower of protons from outer space deposits equal charges +q on the earth and the moon and the electrostatic repulsion then exactly counter balances the gravitational attraction. How large is q ?

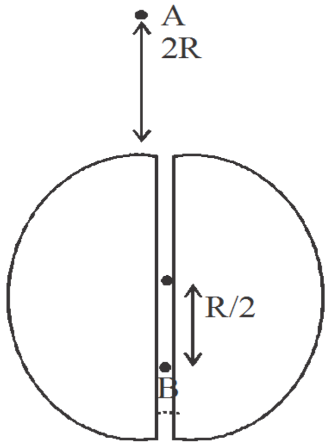
(A) $\sqrt{4\pi\varepsilon_0 GM_eM_m}$

(B) $\sqrt{2\pi\varepsilon_0 GM_eM_m}$

(C) $\sqrt{8\pi\varepsilon_0 GM_eM_m}$

(D) $\sqrt{10\pi\varepsilon_0 GM_eM_m}$

13) Suppose, if a tunnel is dug along the diameter of the earth and a body of mass m is released from a point 'A' at a distance 2R above earth along the line of tunnel. If M is the mass of the earth & R is the radius of the earth. The velocity of the body during its fall when it crosses the point B at a



distance $\frac{R}{2}$ below the earth centre as shown, is

(A) $\frac{5}{2}\sqrt{\frac{GM}{3R}}$

(B) $\sqrt{\frac{GM}{3R}}$

(C) $2\sqrt{\frac{GM}{3R}}$

(D) $\frac{3}{2}\sqrt{\frac{GM}{3R}}$

14) Suppose the gravitational force varies inversely as the nth power of distance. Then, the time period of a planet in circular orbit of radius R around the sun will be proportional to

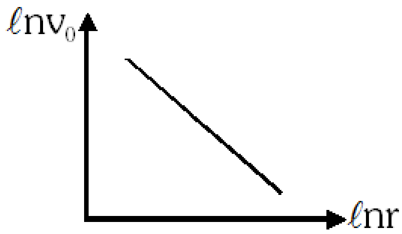
(A) R^{n}

(B) $R^{\frac{n+1}{2}}$

(C) $R^{\frac{n-1}{2}}$

(D) R^{-n}

15) If the law of gravitation be such that the force of attraction between two particles vary inversely as the 5/2th power of their separation, then the graph of orbital velocity v_0 plotted against the distance r of a satellite from the earth's centre on a log-log scale is shown alongside. The slope of line will be-



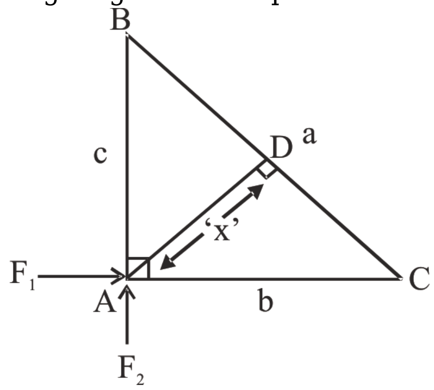
(A) $-\frac{5}{4}$

(B) $-\frac{5}{2}$

(C) $-\frac{3}{4}$

(D) -1

16) Two forces acting on point A on their side and having magnitude reciprocal to length of side



then resultant of these forces will be proportional to:

(A) $\frac{1}{BC}$

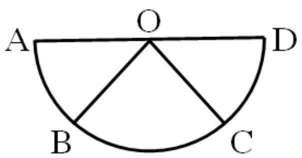
(B) $\frac{1}{AD}$

(C) $\frac{1}{BD}$

(D) $\frac{1}{CD}$

17) **Assertion A :** If A, B, C, D are four points on a semi-circular arc with centre at 'O' such that $|\overrightarrow{AB}| = |\overrightarrow{BC}| = |\overrightarrow{CD}|$ then $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} = 4\,\overrightarrow{AO} + \overrightarrow{OB} + \overrightarrow{OC}$

Reason R : Polygon law of vector addition yields $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{AD} = 2\,\overrightarrow{AO}$



In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) **A** is correct but **R** is not correct.

(B) **A** is not correct but **R** is correct.

(C) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(D) Both **A** and **R** are correct but **R** is not the correct explanation of **A**.

18) Three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are described as follows;

Vector	Magnitude	Direction
$\vec{a}$	5	North
$\vec{b}$	5	East
$\vec{c}$	c	Vertically upwards

Find c such that $\left(\vec{a} + \vec{b} + \vec{c}\right)$ makes 37° with upward vertical

(A) $\frac{5\sqrt{2}}{4}$

(B) $\frac{18\sqrt{2}}{3}$

(C) $5\sqrt{2}$

(D) $\frac{20\sqrt{2}}{3}$

19) In SI units, the dimensions of $\sqrt{\frac{\epsilon_0}{\mu_0}}$ is :

(A) $A^{-1}\,TML^{3}$

(B) $A^{2}T^{3}M^{-1}L^{-2}$

(C) $AT^{2}M^{-1}L^{-1}$

(D) $AT^{-3}ML^{3/2}$

20) If surface tension (S), Moment of inertia (I) and Planck's constant (h), were to be taken as the fundamental units, the dimensional formula for linear momentum would be :-

(A) $S^{3/2}I^{1/2}h^{0}$

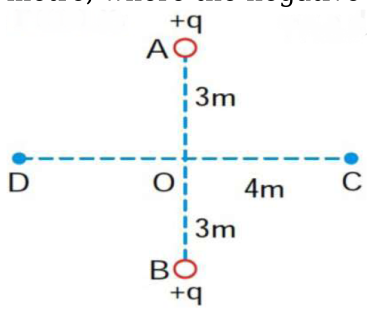
(B) $S^{1/2}I^{1/2}h^{0}$

(C) $S^{1/2}I^{1/2}h^{-1}$

(D) $S^{1/2}I^{3/2}h^{-1}$

SECTION-B

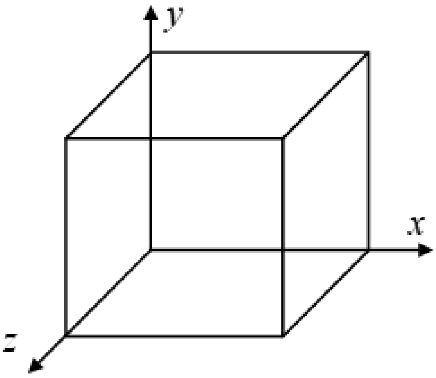
1) Two fixed, equal, positive charges, each of magnitude 5×10^{-5}C are located at points A and B separated by a distance of 6m. An equal and opposite charge moves towards them along the line COD, the perpendicular bisector of the line AB. The moving charge when it reaches the point C at a distance of 4m from O, has a kinetic energy of 4 joules. The distance of the farthest point DO, in metre, where the negative charge will reach before returning towards C is $6\sqrt{x}$. Find x.



2) Electric potential in volt (in region) is given by $V = 6x + 8y + 4z^2$ Calculate the electric force (in N) acting on 2C point charge placed at origin.

3) A spherical cavity of radius 2m is made whose centre is at a distance of 2m from centre of a uniformly charged solid sphere of density r = 8.85×10^{-12} C/m^3 and radius 5m. If electric field inside cavity is given by x/y N/C, fill value of (y-x). (y & x are minimum possible integers).

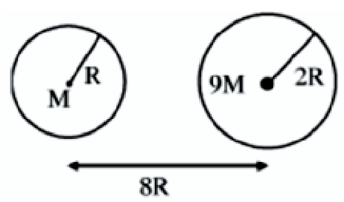
4) An electric field given by $\vec{E} = 4\hat{i} - 3(y^2 + 2)\hat{j}$ pierces Gaussian's cube of side 1*m* placed at origin such that its three sides represents *x*, *y* and *z* axes. The net charge enclosed within the cube is given by $-n\varepsilon_0$. Find the value of *n*.



5) A particle of mass m_0 is projected vertically upward from surface of earth with a speed of $\sqrt{\frac{5GM_e}{4R}}$. The maximum height of the particle from the earth surface is $\frac{kR}{3}$. [Mass of the earth M_e and R is radius of earth]. What is the value of k.

6) A particle is projected radially outwards from the surface of a planet with an initial speed of 4.0 km/s. Maximum height attained by the particle is h = 900 km from the planet's surface. Find the radius R (in km) of the planet taking g = 10 m/s^2 on the surface of the planet. Fill R/900 in OMR sheet.

7) Suppose two planets (spherical in shape) of radii R and 2R, but mass M and 9 M respectively have a centre to centre separation 8 R as shown in the figure. A satellite of mass 'm' is projected from the surface of the planet of mass 'M' directly towards the centre of the second planet. The minimum speed 'v' required for the satellite to reach the surface of the second planet is $\sqrt{\frac{a}{7}\frac{GM}{R}}$ (Where the fraction is in lowest form) then the value of 'a' is ....
[Given : The two planets are fixed in their position]



8) If the expression for the gravitational field due to a uniform rod of length L and mass M at a point on its perpendicular bisector at a distance d from the centre is given by $\frac{aGm}{d\sqrt{L^2 + bd^2}}$, Then find the value of ab.

9) If the earth suddenly shrinks to $\frac{1}{64}^{th}$ of its original volume with its mass remaining the same, the

period of rotation of earth becomes $\frac{24}{x}h$. The value of x is ________ .

10) A satellite is launched in the equatorial plane in such a way that it can transmit signals upto 60° latitude on the earth. The orbital velocity of the satellite is found to be $\sqrt{\frac{GM}{\alpha R}}$. Find the value of α.

PART 2 : CHEMISTRY

SECTION-A

1) A polystyrene having formula $Br_3C_6H_3(C_3H_8)_n$ found to contain 10.46% of bromine by weight. The value of n :- (at. wt of Br = 80)

(A) 40
(B) 43
(C) 45
(D) 47

2) 60 gm of an organic compound containing C,H & O atoms on complete combustion gave 88gm CO_2 & 36gm H_2O. The empirical formula of the organic compound is :

(A) C_2HO
(B) CHO
(C) CH_2O
(D) CHO_2

3) Select the correct statement(s) for $(NH_4)_3PO_4$:
(i) Ratio of number of oxygen atom to number of hydrogen atom is 1 : 3.
(ii) Ratio of number of cation to number of anion is 3: 1.
(iii) Ratio of number of nitrogen atom to number of oxygen atom is 3 : 4.
(iv) Total number of atoms in one mole of $(NH_4)_3PO_4$ is 20 moles.

(A) (i), (ii)
(B) (i), (ii), (iii)
(C) (ii), (iii)
(D) All of these

4) A 6.90 M solution of KOH in water has 30% of KOH by weight. The density of solution is :

(A) 3.88 g/ml
(B) 13.88 g/ml
(C) 1.4 g/ml
(D) 1.288 g/ml

5) Suppose two elements X and Y combine to form two compounds XY_2 and X_2Y_3. If 0.05 mole of XY_2 weighs 5 g while 3.011×10^{23} molecules of X_2Y_3 weighs 85 g, then atomic masses of X and Y are respectively :

(A) 20, 30

(B) 30, 40

(C) 40, 30

(D) 80, 60

6) If 0.5 moles of $BaCl_2$ is mixed with 0.2 moles of Na_3PO_4, the maximum amount of $Ba_3(PO_4)_2$ that can be formed is:

(A) 0.7 mol

(B) 0.5 mol

(C) 0.2 mol

(D) 0.1 mol

7) A solution is made by dissolving $CaBr_2$ in water (solvent) such that mass fraction of solute and solvent is same in the solution. The molality of solution is: (Atomic mass Br = 80, Ca = 40)

(A) 2.5 m

(B) 55.55 m

(C) 2 m

(D) 5 m

8) The volume of one mole of water of 277K is 18ml. One ml of water contains 20 drops. The number of molecules in one drop of water will be :

(A) 1.07×10^{21}

(B) 1.67×10^{21}

(C) 2.67×10^{21}

(D) 1.67×10^{20}

9) The % by volume of C_4H_{10} in a gaseous mixture of C_4H_{10}, CH_4 and CO is 40. When 200 ml of the mixture is burnt in excess of O_2. Find volume (in ml) of CO_2 produced.

(A) 220

(B) 340

(C) 440

(D) 560

10) 20 mL of a gaseous hydrocarbon was exploded with 120 mL of oxygen. A contraction of 60 mL was observed, and a further contraction of 60 mL took place when KOH was added. What is the formula of the hydrocarbon :

(A) C_3H_6

(B) C_3H_8

(C) C_2H_6

(D) C_4H_{10}

11) In which of the following, the oxidation number of oxygen has been arranged in increasing order?

(A) $BaO_2 < KO_2 < O_3 < OF_2$

(B) $OF_2 < KO_2 < BaO_2 < O_3$

(C) $BaO_2 < O_3 < OF_2 < KO_2$

(D) $KO_2 < OF_2 < O_3 < BaO_2$

12) How many moles of MnO_4^- will react with 1 mole of ferrous oxalate ion in acidic medium.

(A) $\frac{1}{5}$

(B) $\frac{2}{5}$

(C) $\frac{3}{5}$

(D) $\frac{5}{3}$

13) One mole of P_2H_4 loses ten moles of electrons to form a new compound Y. Assuming that all the phosphorous appears in the new compound, what is the oxidation state of phosphorous in Y ?

(A) –1

(B) –3

(C) +3

(D) +5

14) In which of the following reactions H_2O_2 acts as a reducing agent?

(i) $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$

(ii) $H_2O_2 - 2e^- \rightarrow O_2 + 2H^+$

(iii) $H_2O_2 + 2e^- \rightarrow 2OH^-$

(iv) $H_2O_2 + 2OH^- - 2e^- \rightarrow O_2 + 2H_2O$

(A) (i), (ii)

(B) (iii), (iv)

(C) (i), (iii)

(D) (ii), (iv)

15) A_2O_x is oxidised to AO_3^- by MnO_4^- in acidic medium. If 1.5×10^{-3} mole of A_2O_x requires 40 ml of 0.03M $KMnO_4$ solution in acidic medium. Which of the following statement(s) is/are correct?

(A) The value of "x" = 1

(B) The value of "x" = 3

(C) Empirical formula of oxide is AO_3.

(D) Empirical formula of oxide is A_2O.

16) How many moles of stannous oxalate can be oxidised into Sn^{+4} and CO_2 using 1 mole of $K_2Cr_2O_7$:-

(A) 0.5

(B) 1

(C) 1.5

(D) 2

17) Amount of oxalic acid present in a solution can be determined by its titration with $KMnO_4$ solution in the presence of H_2SO_4. The titration gives unsatisfactory result when carried out in the presence of HCl, because HCl :

(A) furnishes H^+ ions in addition to those from oxalic acid.

(B) reduces permanganate to Mn^{2+}.

(C) oxidises oxalic acid to carbon dioxide and water.

(D) gets oxidised by oxalic acid to chlorine.

18) Calculate the mass of anhydrous oxalic acid. Which can be oxidised to $CO_{2(g)}$ by 100 ml of an MnO_4^- solution, 10 ml of which is capable of oxidising 50 ml of 1 N I^- to I_2.

(A) 45 gm

(B) 22.5 gm

(C) 30 gm

(D) 12.25 gm

19) Moles of $K_2Cr_2O_7$ used to oxidise 1 mol $Fe_{0.92}O$ to Fe^{+3} are :-

(A) $\frac{0.92}{6}$

(B) $\frac{70}{92} \times \frac{1}{6}$

(C) $\frac{0.76}{6}$

(D) $\frac{70}{92} \times \frac{1}{3}$

20) Given below are two statements :
Statement-I: S_8 solid undergoes disproportionation reaction under alkaline conditions to form S^{2-} and $S_2O_3^{2-}$
Statement-II: ClO_4^- can undergo disproportionation reaction under acidic condition.
In the light of the above statements, choose the ***most appropriate answer*** from the options given below :

(A) Statement I is correct but statement II is incorrect.

(B) Statement I is incorrect but statement II is correct

(C) Both statement I and statement II are incorrect

(D) Both statement I and statement II are correct

SECTION-B

1) One litre of a sample of hard water contains 11.1 mg of $CaCl_2$ & 9.5 mg of $MgCl_2$. What is degree of hardness in terms of ppm of $CaCO_3$?

2) Balance the following equation and find the sum of the co-efficients of reactants and products.
__$KMnO_4$ + __H_2O_2 + __H_2SO_4 $\rightarrow$ __$MnSO_4$ + __O_2 + __H_2O + __K_2SO_4.

3) If 50 mL of 0.5 M oxalic acid is required to neutralise 25 mL of NaOH solution, the amount of NaOH in 50 mL of given NaOH solution is ____ g.

4) Normality of 2M H_3PO_3 solution is xN. The value of x is ______.

5) 15 mL of aqueous solution of Fe^{2+} in acidic medium completely reacted with 20 mL of 0.03 M aqueous $Cr_2O_7^{2-}$. The molarity of the Fe^{2+} solution is _____$\times 10^{-2}$M (Round off to the Nearest Integer).

6) Consider the following reaction:
$3PbCl_2 + 2(NH_4)_3PO_4 \rightarrow Pb_3(PO_4)_2 + 6NH_4Cl$
If 72 mmol of $PbCl_2$ is mixed with 50 mmol of $(NH_4)_3PO_4$, then amount of $Pb_3(PO_4)_2$ formed is ...... mmol. (nearest integer)

7) A sample of H_2SO_4 is 40% by mass and shows density 1.47 g/ml. What is the molarity of the acid?

8) A chemist wants to prepare diborane by the reaction $6LiH + 8BF_3 \rightarrow 6LiBF_4 + B_2H_6$.
If we start with 32.0 moles of each of LiH and BF_3. How many moles of B_2H_6 can be prepared.

9) 10 mL of gaseous hydrocarbon on combustion gives 40 mL of CO_2(g) and 50 mL of water vapour. Total number of carbon and hydrogen atoms in the hydrocarbon is _____.

10) The molarity of 1L orthophosphoric acid (H_3PO_4) having 70% purity by weight (specific gravity 1.54 g cm^{-3}) is ______M.
(Molar mass of $H_3PO_4 = 98 \text{ g mol}^{-1}$)

PART 3 : MATHEMATICS

SECTION-A

1) If a, b, c be the sides of a triangle where $a \neq b \neq c$ then the roots of the equation $x^2 + (a + b + c)x + ab + bc + ca = 0$ are

(A) real and integers

(B) imaginary

(C) real and irrational

(D) None of these

2) The equations $x^3 + 5x^2 + px + q = 0$ and $x^3 + 7x^2 + px + r = 0$ have two roots in common. If the third root of each equation is represented by x_1 and x_2 respectively, then the ordered pair (x_1, x_2) is :

(A) (– 5, – 7)

(B) (1, – 1)

(C) (–1, 1)

(D) (5, 7)

3) Let 'α' and 'β' are the roots of quadratic $x^2 - 5x - 1 = 0$ then the value of $\dfrac{\alpha^{15} + \alpha^{11} + \beta^{15} + \beta^{11}}{\alpha^{13} + \beta^{13}}$ is equal to :

(A) 7

(B) 17

(C) 27

(D) 23

4) If α and β be the roots of $x^2 - 3x + 5 = 0$ then the quadratic equation whose roots are $\alpha^3 - \alpha^2 - \alpha + 12$ and $\beta^3 + 2\beta^2 - 10\beta + 31$ is

(A) $x^2 + 7x + 10 = 0$

(B) $x^2 + 8x + 12 = 0$

(C) $x^2 - 7x + 10 = 0$

(D) $x^2 - 8x + 12 = 0$

5) If α, β are roots of the equation $x^2 - 2mx + m^2 - 1 = 0$ then the number of integral values of m for which $\alpha, \beta \in (-2, 4)$ is

(A) 0

(B) 1

(C) 2

(D) 3

6) Let $p(x) = a_0 + a_1x + \ldots + a_nx^n$ be a polynomial with integer coefficient and $a_i \in \{0, 1\}$ for all i = 0, 1, 2, ... n given $p\left(\sqrt{2}\right) = 13 + 19\sqrt{2}$ then p(1) is

(A) 4

(B) 5
(C) 6
(D) 8

7) Let α, β, γ be the roots of $(x - a)(x - b)(x - c) = d$, $d \neq 0$ then the roots of the equation $(x - \alpha)(x - \beta)(x - \gamma) + d = 0$ are

(A) $a + 1, b + 1, c + 1$
(B) a, b, c
(C) $a - 1, b - 1, c - 1$
(D) $\frac{a}{b}, \frac{b}{c}, \frac{c}{a}$

8) If the reciprocals of the roots of the equation $10x^3 - ax^2 - 54x - 27 = 0$ are in A.P., then the value of a is :

(A) 6
(B) 8
(C) –9
(D) 9

9) Let P(S) denote the power set of S={1,2,3, ..., 10}. Define the relations R_1 and R_2 on P(S) as AR_1B if $(A \cap B^c) \cup (B \cap A^c) =$ ▯ and AR_2B if $A \cup B^c = B \cup A^c$, $\forall$ A, B $\in$ P(S). Then :

(A) both R_1 and R_2 are equivalence relations
(B) only R_1 is an equivalence relation
(C) only R_2 is an equivalence relation
(D) both R_1 and R_2 are not equivalence relations

10) Two finite sets have m and n elements. The number of subsets of the first set is 112 more than that of the second set. The absolute difference of values of m and n is

(A) 2
(B) 3
(C) 4
(D) 5

11) Let $S_1 = \{5, 7, 9, 11\}$ and $S_2 = \{2, 4, 6, 8.......32\}$. If $M = \{a_1 + a_2 + ... + a_k : k \in N \, \forall \, a_1, a_2, ...a_k \in S_1\}$ then the number of elements in set $S_2 - M$ is equal to

(A) 6
(B) 8
(C) 10
(D) 12

12) Let R_1 and R_2 be relations on the set {1,2,3,......,100} such that
$R_1 = \{(p, p^n) : p$ is a prime and $n \geqslant 0$ is an integer} and
$R_2 = \{(p, p^n) : p$ is a prime and n = 0 or 1 or 2}. Then the number of elements in $R_1 \Delta R_2$ is (where $R_1 \Delta R_2$ denotes symmetric difference of sets)

(A) 10
(B) 6
(C) 5
(D) 4

13) If $2n(A \setminus B) = n(B \setminus A)$ and $5n(A \cap B) = n(A) + 3n(B)$, where $P \setminus Q = P \cap Q^c$. If $n(A \cup B) \leqslant 10$, then the value of $\frac{n(A).n(B).n(A \cap B)}{8}$ is -
(where n(X) denotes the cardinal number of set X)

(A) 63
(B) 72
(C) 90
(D) 70

14) In a group of 60 person 40 speak English and 30 speak Hindi. Each person speaks at least one of the two languages. If the number of persons who speak only Hindi is α and the number of person who speaks both the languages is β, then find the equation of line whose x-intercept is α and y-intercept is β.

(A) x + 2y = 20
(B) x - 2y = 20
(C) 2x + y = 10
(D) 2x - y = 10

15) If a is even natural number, b is odd natural number and
$$\left[\frac{a}{b}\right] + \left[\frac{2a}{b}\right] + \left[\frac{3a}{b}\right] + \ldots + \left[\frac{(b-1)a}{b}\right] = \frac{(a-k_1)(b-1)}{k_2}$$
(where [x] represents greatest integer function of x), then $k_1 + k_2$ equals

(A) 2
(B) 3
(C) 4
(D) 5

16) Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \ln(x + \sqrt{x^2 + 1})$, then number of solutions of $|f^{-1}(x)| = e^{-|x|}$ is :-

(A) 1
(B) 2

(C) 3

(D) Infinite

17) Which of the following functions is an odd function ?
(where sgn(x) denotes signum function of x)

(A) $|x - 2| + (x + 2)\,\mathrm{sgn}(x + 2)$

(B) $\dfrac{1}{x(e^x - 1)} + \dfrac{1}{2x}$

(C) $\log\left(\sin x + \sqrt{1 + \sin^2 x}\right)$

(D) $e^{-4x}\left(e^{2x} - 1\right)^4$

18) If the domain of the function $f(x) = \dfrac{\sqrt{x^2 - 25}}{(4 - x^2)} + \log_{10}(x^2 + 2x - 15)$ is $(-\infty, \alpha) \cup [\beta, \infty)$, then $\alpha^2 + \beta^3$ is equal to :

(A) 140

(B) 175

(C) 150

(D) 125

19) If f(x) be a continuous function on R such that $f(x) = f(8 - x)$ and $f(x+10) = f(14 - x)$, $\forall x \in R$ and $f(0) = f(3) = 6$ then minimum number of 'α' such that $f(\alpha) = 6$, where $\alpha \in [0, 100]$

(A) 26

(B) 24

(C) 16

(D) 20

20) A function has the property
(i) $f(1) = 1$
(ii) $f(3x) = 9f(x) + 8$
(iii) $f(x + 4) = f(x) + 16x + 32$,
then the value of f(15) is

(A) 899

(B) 219

(C) 449

(D) 53

SECTION-B

1) Let $P(x) = x^2 + bx + c$ where $b, c \in R$. If $P(P(1)) = 0$, $P(P(2)) = 0$ and $P(1) \neq P(2)$ then the absolute value of $(b + c)$ is equal to

2) Let α and β be the positive solutions of the quadratic equation $x^2 - 1154x + 1 = 0$, then the value of $\sqrt[4]{\alpha} + \sqrt[4]{\beta}$ is equal to ________

3) Let α,β and γ be roots of $x^3 + 6x^2 - 2x + 1 = 0$ and $S_n = \alpha^n + \beta^n + \gamma^n \ \forall\ n \in N$, then value of $\dfrac{S_{14} + S_{11}}{S_{12} - 3S_{13}}$ is

4) Let α, β be roots of quadratic equation $x^2 + (5 - a)x + a = 0$. Given a_1, a_2, a_3 ($a_3 > a_2 > a_1$) are three values of a satisfying, $\alpha\beta^3 + \alpha^3\beta = 14(2 - a)$. If $4a_1(x^2 + 1) + p.a_2(x + 1) - a_3 > 0$ is true for every $x \in R$, then number of possible integral values in range of p is equal to

5) The set of all real parameter 'a' for which the equation $x^4 - 2ax^2 + x + a^2 - a = 0$ has all real solutions, is given by $\left[\dfrac{m}{n}, \infty\right)$ where m and n are relatively prime positive integers, find the value of (m + n).

6) If [x] + [2x] + [3x] + ... + [2020x] = 1009□, where $x = \dfrac{1010}{2021}$, then sum of digits of □, is (Where [.] denotes greatest integer function)

7) If the periodic function f (x) satisfies the equation $f(x + 1) + f(x - 1) = \sqrt{3}f(x), \forall x \in R$ and the period of f (x) is 4λ then λ = ___.

8) The number of integral values of m for which $f : R \to R; f(x) = \dfrac{x^3}{3} + (m - 1)x^2 + (m + 5)x + n$ is bijective is______.

9) If $f(x) = \sqrt{\sqrt{38 - 4x^2 - 11x}.(x^2 - 2x - 15)}$ and sum of integers which are in domain of f(x) is 's' then the value of |2- s| is

10) Let ƒ : [-2,2] → R, where $g(x) = x^{2015} + \text{sgn}(x) + \left[\dfrac{x^2 + 1}{p}\right]$ be an odd function for all x ∈ [-2,2], then the smallest integral value of p is (where [.] denotes greatest integer function and sgn(.) denotes signum function)

ANSWER KEYS

PART 1 : PHYSICS

SACTION-A

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	D	B	B	D	D	B	A	B	D	C	B	A	A	B	C	B	D	D	B	B

SECTION-B

Q.	21	22	23	24	25	26	27	28	29	30
A.	2	20	1	3	5	8	4	8	16	2

PART 2 : CHEMISTRY

SECTION-A

Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
A.	C	C	D	D	C	D	D	B	C	B	A	C	C	D	B	C	B	B	C	A

SECTION-B

Q.	51	52	53	54	55	56	57	58	59	60
A.	20	26	4	4	24	24	6	4	14	11

PART 3 : MATHEMATICS

SECTION-A

Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	B	A	C	D	D	C	B	D	A	B	C	B	A	A	B	B	C	C	A	C

SECTION-B

Q.	81	82	83	84	85	86	87	88	89	90
A.	2	6	2	1	7	2	3	6	7	6

SOLUTIONS

PART 1 : PHYSICS

1) $V_A \propto \rho L^2$

Divide cube into smaller cube of length $\frac{L}{2}$
Therefore

$$V_C \propto 8 \cdot (\rho)\left(\frac{L}{2}\right)^2 \propto 2\rho L^2$$

$V_C \propto 2\rho L^2$

$$\frac{V_A}{V_C} = \frac{1}{2}$$

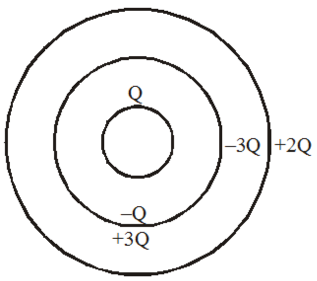
2)

Only half of field lines would pass.

3) $$F = \frac{\left(9 \times 10^9\right)\left(100 \times 10^{-6}\right)\left(4 \times 10^{-6}\right)}{(20)^3}$$

$$\left(6\sqrt{3}\hat{i} - 8\sqrt{3}\hat{j} + 10\hat{k}\right)$$

4)



5)

$$\oint \vec{E}.d\vec{A} = \frac{q_{in}}{\varepsilon_0}$$

Gauss Law is applicable in both the below cases.

$$\oint \vec{E}.d\vec{A} = 0, \oint \vec{E}.d\vec{A} \neq 0$$

6) Conceptual

7) Conceptual

8) Conceptual

9)

From Kepler's second law we can say that areal velocity is constant
Let Area SAB = A
Then Area SCD = 2A
Let Areal velocity = V = Constant
Then $v \times t_1 = 2A$
$v \times t_2 = A$
$\frac{t_1}{t_2} = 2$
$t_1 = 2t_2$

10)

$g = \frac{GM}{R^2}$
mass & radius are same

11)

$V_e = \sqrt{\frac{2GM}{R}}$
$\frac{M_1}{R_1} = \frac{M_2}{R_2}$
$M_1R_2 = M_2R_1$
Hence reason R is not correct.

12)

Here, electrostatic repulsive force = gravitational attractive force
i.e., $\frac{1}{4\pi\varepsilon_0} \cdot \frac{q^2}{r^2} = \frac{GM_eM_m}{r^2}$ or $q = \sqrt{4\pi\varepsilon_0 GM_eM_m}$

13)

$V_A = \frac{-GM}{3R}$,
$V_B = \frac{-GM}{2R^3}\left(3R^2 - \frac{R^2}{4}\right) = \frac{-11GM}{8R}$
By conservation of energy
$$mV_A = mV_B + \frac{1}{2}m\nu^2$$
$$\Rightarrow \quad \frac{-GMm}{3R} = \frac{-11GMm}{8R} + \frac{1}{2}m\nu^2$$
$$\Rightarrow \quad \nu = \frac{5}{2}\sqrt{\frac{GM}{3R}}$$

14)

$$F = \frac{GMm}{R^n} = m\omega^2 R$$

$$\omega = \sqrt{\frac{GM}{R^{n+1}}}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{R^{n+1}}{GM}}$$

$$T = R^{\frac{n+1}{2}}$$

15)

$$\frac{mv_0^2}{r} = \frac{GMm}{r^{5/2}} \Rightarrow v_0 = \frac{\sqrt{GM}}{r^{3/4}}$$

$$\Rightarrow \ell n v_0 = \ell n\sqrt{GM} - \frac{3}{4}\ell n r$$

16)

$$\vec{F}_1 = \frac{1}{|AC|} = \frac{1}{b}$$

$$F_2 = \frac{1}{|AB|} = \frac{1}{c}$$

Area of $\Delta = \frac{1}{2} \times bc = \frac{1}{2} \times a \times x$

$$x = \frac{bc}{a}$$

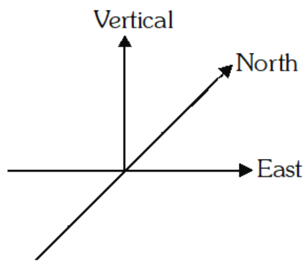
$$F_{net} = \frac{a}{bc} = \frac{1}{x}$$

$$F_{net} = \frac{1}{|AD|}$$

17)

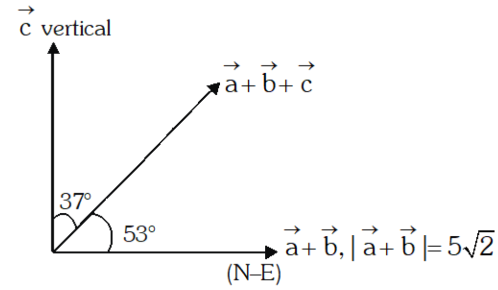
Polygon law is applicable in both but the equation given in the reason is not useful in explaining the assertion.

18)



$$\left|\vec{a} + \vec{b}\right| = 5\sqrt{2}$$

$\vec{a} + \vec{b}$ is along N – E.



tan 53° = $\frac{|\vec{c}|}{|\vec{a}+\vec{b}|}$

$\frac{4}{3} = \frac{c}{5\sqrt{2}}$

c = $\frac{20\sqrt{2}}{3}$

19)

dimension of $\sqrt{\frac{\varepsilon_0}{\mu_0}}$

$[\varepsilon_0] = [M^{-1}L^{-3}T^4A^2]$

$[\mu_0] = [MLT^{-2}A^{-2}]$

dimensions of $\sqrt{\frac{\varepsilon_0}{\mu_0}} = \left[\frac{M^{-1}L^{-3}T^4A^2}{MLT^{-2}A^{-2}}\right]^{\frac{1}{2}}$

$= [M^{-2}L^{-4}T^6A^4]^{1/2}$

$= [M^{-1}L^{-2}T_3A^2]$

20)

$p = k\, s^a I^b h^c$

where k is dimensionless constant

$MLT^{-1} = (MT^{-2})^a(ML^2)^b(ML^2T^{-1})^c$

a + b + c = 1

2 b + 2c = 1

-2a - c = -1

a = $\frac{1}{2}$ b = $\frac{1}{2}$ c = 0

$s^{1/2}I^{1/2}h^0$

21)

From figure $AC = \sqrt{(AO)^2 + (OC)^2}$

$AC = \sqrt{3^2 + 4^2} = 5m$ Similarly, BC = 5m

P.E. at C is $U_C = 2 \times 9 \times 10^9 = \frac{(q) \times (-q)}{AC}$

K.E. at C is 4J

Total energy at C is $E_C = P.E + K.E$

$\Rightarrow E_C = -9 + 4 = -5J$

Potential energy at D is U_D, given by

$$U_D = \frac{-2 \times 9 \times 10^9 \left(5 \times 10^{-5}\right)^2}{r}$$

where AD = BD = r (say)

$\Rightarrow U_D = \frac{-45}{r} J$ K.E at D = 0

So, total energy at D is

$$E_D = 0 + \frac{-45}{r} = \frac{-45}{r} J$$

By Law of Conversation of Energy,

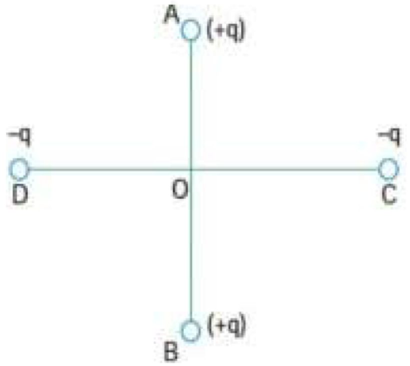
$$U_C + K_C = U_D + K_D$$

$$\Rightarrow \frac{-45}{R} = -5; \Rightarrow r = 9m$$

Now $OD = \sqrt{(AD)^2 - (AO)^2}$

$$\Rightarrow OD = \sqrt{9^2 - 3^2} = \sqrt{72} m$$

$$\Rightarrow OD = 6\sqrt{2} \Rightarrow x = 2$$



22)

$$E_x = -\frac{\partial v}{\partial x} = -6\hat{i}$$

$$E_y = -\frac{\partial v}{\partial y} = -8\hat{j}$$

$$E_z = -\frac{\partial v}{\partial z} = -8\hat{k}$$

$\vec{E}_{net}$ at origin $= \sqrt{6^2 + 8^2} = 10$

$\Rightarrow |\vec{F}| = |q\vec{E}| = 20N$

23)

Electric field inside spherical cavity $E = \frac{\rho(OO')}{3\varepsilon_0}$

where OO' = distance between centre of sphere & cavity

$$E = \frac{2}{3} \frac{N}{C}$$

$\Rightarrow$ y = 3, x = 2

24)

Net flux in x-direction = 0 ;

Net flux in y-direction = $A\left[3\left(1^2+2\right)\right]-A\left[3\left(0^2+2\right)\right]$ $\Rightarrow q = -3\epsilon^0$ (as A = 1m²)

25)

by energy conservation

$$\frac{1}{2}m\left(\frac{5GM}{4R}\right)+\left(\frac{-GMm}{R}\right)=O-\frac{GMm}{(R+h)}$$

$$\Rightarrow h=\frac{5R}{3}$$

26) $\frac{1}{2}mv^2-\frac{GMm}{R}=-\frac{GMm}{R+h}$

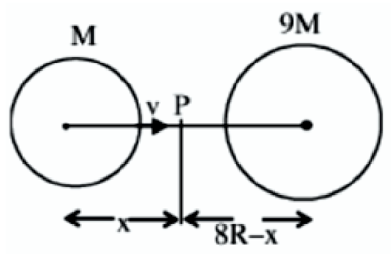
$$\frac{v^2}{2}=\frac{GMR}{R(R+h)}=\frac{GM}{R}\frac{Rh}{R+h}$$

$$\frac{v^2}{2}=g\times\frac{Rh}{R+h}$$

$(R + 900 \times 10^3) \times 38 \times 10^6 = 10 \times R \times 900 \times 10^3$

$R = 72 \times 10^5$ m $\Rightarrow$ 7200 km $\Rightarrow \frac{R}{900}=8$

27)



Acceleration due to gravity will be zero at P therefore,

$$\frac{GM}{x^2}=\frac{G9M}{(8R-x)^2}$$

8R - x = 3x

x = 2R

Apply conservation of energy and consider velocity at P is zero.

$$\frac{1}{2}mv^2-\frac{GMm}{R}-\frac{G9Mm}{7R}=0-\frac{GMm}{2R}-\frac{G9Mm}{6R}$$

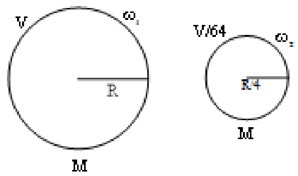
$$\therefore V=\sqrt{\frac{4}{7}\frac{GM}{R}}$$

28)

$$\frac{2Gm}{d\sqrt{L^2+4d^2}}$$

29)

From conservation of angular momentum



$$\frac{2}{5}MR^2\omega_1 = \frac{2}{5}M\left(\frac{R}{4}\right)^2\omega_2$$

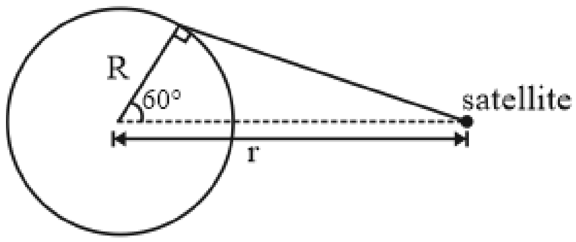
$$\Rightarrow MR^2\omega_1 = \frac{MR^2}{16}\omega_2$$

$$\Rightarrow \frac{\omega_1}{\omega_2} = \frac{1}{16} \Rightarrow \frac{T_2}{T_1} = \frac{1}{16} \Rightarrow \frac{T_1}{T_2} = \frac{16}{1} = \frac{t_1}{x}$$

$$\therefore t_1 = 24 \Rightarrow \frac{16}{1} = \frac{24}{t_2} \Rightarrow x = 16$$

30)

$r\cos 60^\circ = R \Rightarrow r = 2R$ orbital velocity $v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{2R}} \Rightarrow \alpha = 2$



PART 2 : CHEMISTRY

31)

Let M is molecular weight

moles of Br = $\frac{100}{M} \times 3$

wt. of Br = $\frac{100}{M} \times 3 \times 80 = 10.46$

M = 2295.45 = 240 + 75 + 44n

n = 45

32)

$C_xH_yO_z + O_2 \rightarrow xCO_2 + \frac{y}{z}H_2O$

60 gm	88 gm	36 gm
	2 moles	2 moles

x × moles of $C_xH_yO_z$ = 1 × 2

y × moles of $C_xH_yO_z$ = 2 × 2

$$\frac{x}{y} = \frac{1}{2}$$

33)

(1) Oxygen atom : Hydrogen atom
4 12 [1 : 3]
(2) $3NH_4^+$: $1PO_4^{3-}$ [3 : 1]
(3) N atom : Oxygen atom [3 : 4]
(4) $(NH_4)_3PO_4$ | 3N + 12H + P + 4 O
1 mole
1 mole of $(NH_4)_3PO_4$ having 20-Mole of atoms and or $(20 \times N_A)$ atoms.

34)

Weight of KOH in 1 litre solution = $6.9 \times 56 = \frac{30}{100} \times$ weight of solution

 weight of solution = $\frac{6.9 \times 56}{0.3}$ = 1288g

 density of solution = $\frac{1288}{1000}$ g/ml = 1.288g/ml

35) wt of 0.05 mole xy_2 = 5g

wt of 1 mole $xy_2 = \frac{5}{.05}$ =100 g
x + 2y = 100 ..(1)
moles of $x_2y_3 = \frac{3.011 \times 10}{6.02 \times 10^{23}}$ = .5 mol
wt of 0.5 mole of x_2y_3 = 85g
wt of 1 mole of $x_2y_3 = \frac{85}{0.5}$ 170g
2x + 3y = 170 ..(2)
from (1) $eq^n \Rightarrow x + 2y = 100$
(2) $eq^n \Rightarrow 2x + 3y = 170$

[x = 40] [y = 30]

36)

Let us first solve this Prob by writing the complete balanced reaction.
$3BaCl_2 + 2\,Na_3PO_4 \rightarrow Ba_3(PO_4)_2 \downarrow + 6NaCl$

We can see that the moles of $BaCl_2$ used are $\frac{3}{2}$ times the moles of Na_3PO_4.

Therefore, to react with 0.2 mol of Na_3PO_4, the moles of $BaCl_2$ required would be $0.2 \times \frac{3}{2} = 0.3$. Since $BaCl_2$ is 0.5 mol, we can conclude that Na_3PO_4 is the limiting reagent. Therefore, moles of $Ba_3(PO_4)_2$ formed is $0.2 \times \frac{1}{2} = 0.1$ mol.

37)

Solute = $CaBr_2$ $\boxed{M_W = 200}$
Solvent = H_2O
Same mass fraction ⇒ mass of solute = mass of solvent

$= x$ gm (say)

Molality $= \dfrac{\text{moles of solute}}{\text{mass of solvent (in kg)}}$

$m = \dfrac{\frac{x}{200}}{\frac{x}{1000}} = 5$

38)

18 ml H_2O = 18g H_2O

18g $H_2O \rightarrow 6 \times 10^{23}$ molecules of H_2O

18×20 drops $\rightarrow 6 \times 10^{23}$ molecules of H_2O

1 drops $\rightarrow$ x molecules of H_2O

$x = \dfrac{6 \times 10^{23}}{18 \times 20} \Rightarrow 1.67 \times 10^{21}$

39) $C_4H_{10} = 200 \times \dfrac{40}{100} = 80\,\text{ml}$

$C_4H_{10} + \frac{13}{2}O_2 \rightarrow 4CO_2 + 5H_2O$

80 $\rightarrow$ $4 \times 80 = 320$

$CH_4 + O_2 \rightarrow 4CO_2 + 2H_2O$

$120 - x \rightarrow 120 - x$

$CO + \frac{1}{2}O_2 \rightarrow CO_2$

$x \rightarrow x$

$CO_2 = 120 - x + 320 + x = 440$ ml

40) $C_xH_y + (x + y/4)\ O_2 \rightarrow XCO_2 + \frac{y}{2}H_2O$

t = 0 20 120

t = ft 0 $120 - 20\left(x + \frac{y}{4}\right)$

$20x = 60$

$V_i - V_f = 60$

$y = 8$

41)

$BaO_2 \Rightarrow O_2^{2-}$ -1

$KO_2 \Rightarrow O_2^{-}$ $-\frac{1}{2}$

$O_3 \Rightarrow O$

$OF_2 \Rightarrow +2$

42)

Eq. of MnO_4^- = Eq of FeC_2O_4

$5 \times nKMnO_4 = 3 \times 1$

$nKMnO_4 = \frac{3}{5}$

43) $P_2H_4 \xrightarrow{-10e^-} \underset{\text{[Compound Y]}}{(P_2H_4)^{+10}}$ in compound Y;

$2x + 4 = +10$

$2x = +6$

$\Rightarrow [x = +3]$

44) H_2O_2 acts as reducing agent when it releases electrons (itself gets oxidised and reduces others) i.e. in (ii) & (iv).

45)

$1.5 \times 10^{-3} \times (5 - x) \times 2 = 0.03 \times 5 \times \frac{40}{1000}$

$x = 3$

46)

$\overset{(+2)(+3)}{Sn}\ C_2O_4 \rightarrow Sn^{+4} + \overset{(+4)}{CO_2}$

n-factor = $(+2 \text{to} + 4) = 2e^-$ loss

$(+3 \text{ to } + 4) \times 2 = 2e^-$ loss

n-factor = $4e^-$ loss

$K_2\overset{(+6)}{Cr_2}O_7 \rightarrow Cr^{+3}$(fixed)

n factor = $(+6 \text{ to } +3) \times 2 = 6e^-$ gain

Total e– loss = e– gain

$4e^- \times ? = 6e^- \times 1$ mol

mole of $SnC_2O_4 = \frac{6}{4} = 1.5$

47) HCl reduces MnO_4^- to Mn^{2+} and itself oxidises to Cl_2.

48)

Let normality of $KMnO_4^-$ solution is N

$N \times 10 = 50 \times 1 \Rightarrow N = 5$

equ. of MnO_4^- = equ. of oxalic acid

$5 \times 100 = \frac{W}{E} \times 1000$

$\Rightarrow W = \frac{500}{1000} \times \frac{90}{2} = 22.5\text{gm}$

49) $Fe_{0.92}O + K_2Cr_2O^7 \longrightarrow Fe^{+3} + Cr^{+3}$

$\frac{+200}{92}$ $\quad +6$

eqv. of $F_{0.92}O$ = eqv. of $K_2Cr_2O_7$

$0.92 \times \left(3 - \frac{200}{90}\right) \times 1 = 6 \times n$

$n = \frac{0.76}{6}$

50)

$S_1 : S_8 + 12OH^{\ominus} \rightarrow 4S^{2-} + 2S_2O_3^{2-} + 6H_2O$

$S_2 : ClO_4^{\ominus}$ cannot undergo disproportionation reaction as chlorine is present in it's highest oxidation state.

51)

Moles of $CaCl_2 = \frac{11.1 \times 10^{-3}}{111} = 10^{-4}$

$\Rightarrow$ equivalent moles of $CaCO_3 = 10^{-4}$

Moles of $MgCl_2 = \frac{9.5 \times 10^{-3}}{95} = 10^{-4}$

$\Rightarrow$ equivalent moles of $CaCO_3 = 10^{-4}$

 Total moles of $CaCO_3 = 2 \times 10^{-4}$

$\Rightarrow$ Weight of $CaCO_3 = 2 \times 10^{-4} \times 100 = 2 \times 10^{-2}$ g

Also, weight of solution = 1000 g

 ppm of $CaCO_3 = \frac{\text{weight of } CaCO_3}{\text{weight of solution}} \times 10^6$

$= \frac{2 \times 10^{-2}}{1000} \times 10^{-6} = 20\,\text{ppm}$

52) Balance reaction is $2KMnO_4 + 5H_2O_2 + 3H_2SO_4 \rightarrow 2MnSO_4 + 5O_2 + 8H_2O + 1K_2SO_4$.
Therefore, Sum of stoichiometric co-efficients
= 2 + 5 + 3 + 2 + 5 + 8 + 1 = 26.

53)

Equivalent of Oxalic acid = Equivalents of NaOH

$50 \times 0.5 \times 2 = 25 \times M \times 1$

$M_{NaOH} = 2M$

W_{NaOH} in 50ml $= 2 \times 50 \times 40 \times 10^{-3}$ g = 4g

54)

H_3PO_3 valency factor = 2

$N = M \times n_f$

55)

$m_{eq.}Fe^{2+} = m_{eq.}Cr_2O_7^{2-}$

$M \times 15 \times 1 = 0.03 \times 6 \times 20$

$M = 0.24$

Therefore, x = 24.

56)

Limiting Reagent is $PbCl_2$
mmol of $Pb_3(PO_4)_2$ formed
$= \frac{\text{mmol of } PbCl_2 \text{ reacted}}{3}$
= 24 mmol

57)

$$\text{Molarity(M)} = \frac{\% \times 10 \times d}{\text{Molecular Weight}}$$
$$= \frac{40 \times 10 \times 1.47}{98} = 6\,M$$

58)

$6LiH + 8BF_3 \rightarrow 6LiBF_3 + B_2H_6$
32 32 4
(L.R)
∵ From 8 mole BF_3 → 1 moles B_2H_6 from
∴ From 32 mole BF_3 = $\frac{1}{8} \times 32 = 4$ mol

59)

$\boxed{\begin{matrix}CxHy\\10ml\end{matrix}} + O_2 \rightarrow CO_2 + H_2O$
$CxHy + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$
10x 5y
10x = 40
x = 4
5y = 50
y = 10
C_4H_{10}

60)

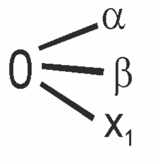
Specific gravity (density) = 1.54 g/cc.
Volume = 1L = 1000 ml
Mass of solution = 1.54×1000
= 1540 g
% purity of H_2SO_4 is 70%
So weight of $H_3PO_4 = 0.7 \times 1540 = 1078$ g
Mole of $H_3PO_4 = \frac{1078}{98} = 11$
Molarity $= \frac{11}{1\,L} = 11$

PART 3 : MATHEMATICS

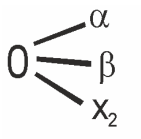
61)

In $\Delta |a - b| < c$; $|b - c| < a$; $|c - a| < b$
$\Rightarrow a^2 + b^2 - 2ab < c^2,\ b^2 + c^2 - 2bc < a^2,$
$c^2 + a^2 - 2ac < b^2$
Hence : $a^2 + b^2 + c^2 < 2(ab + bc + ca)$
$a^2 + b^2 + c^2 - 2(ab + bc + ca) < 0$
D < 0 so roots are imaginary.

62)

$x^3 + 5x^2 + px + q =$ 

$\Rightarrow \alpha + \beta + x_1 = -5,\ \alpha\beta + \beta x_1 + \alpha x_1 = p$ ...(1)

$x^3 + 7x^2 + px + r =$ 

$\Rightarrow \alpha + \beta + x_2 = -7,\ \alpha\beta + \beta x_2 + \alpha x_2 = p$ ...(2)
Subtracting (2) from (1)
$\alpha\beta + \beta x_1 + \alpha x_1 = p$
$$\frac{\alpha\beta + \beta x_2 + \alpha x_2 = p}{\Rightarrow \alpha(x_1 - x_2) + \beta(x_1 - x_2) = 0}$$
$\Rightarrow (x_1 - x_2)(\alpha + \beta) = 0$
$[x_1 \neq x_2]$
$\therefore \alpha + \beta = 0 \Rightarrow x_1 = -5 \Rightarrow x_2 = -7$

63)

$\alpha + \beta = 5$
$\alpha\beta = -1$
$\alpha^{15} + \alpha^{11} \neq \beta^{15} + \beta^{11}$
$= \alpha^{15} + \alpha^{13}\beta^2 + \beta^{15} + \beta^{13}\alpha^2$
$= (\alpha^2 + \beta^2)(\alpha^{13} + \beta^{13})$
Now $\dfrac{\alpha^{15} + \alpha^{11} + \beta^{15} + \beta^{11}}{\alpha^{13} + \beta^{13}} = \alpha^2 + \beta^2$
$= (\alpha + \beta)^2 - 2\alpha\beta$
$= 27$

64) $\alpha^3 - \alpha^2 - \alpha + 12 = (\alpha + 2)(\alpha^2 - 3\alpha + 5) + 2$
$\beta^3 + 2\beta^2 - 10\beta + 31 = (\beta + 5)(\beta^2 - 3\beta + 5) + 6$
So, required equation is $x^2 - 8x + 12 = 0$

65)

$$x = \frac{2m \pm \sqrt{4m^2 - 4(m^2 - 1)}}{2}$$

$= \frac{2m \pm 2}{2} = m + 1 \text{ or } m - 1$

Hence $-2 < m + 1 < 4 \Rightarrow -3 < m < 3$

$-2 < m - 1 < 4 \Rightarrow -1 < m < 5$

Hence $m \in (-1, 3) \Rightarrow m = 0, 1, 2$

$\Rightarrow$ D

66)

$p\left(\sqrt{2}\right) = (a^0 + 2a_2 + 2^2a_4 + \ldots) + \sqrt{2}(a_1 + 2a_3 + 2^2a_5 + \ldots)$

$= 13 + 19\sqrt{2}$

$a_0 + 2a_2 + 2^2a_4 + \ldots = 13$

$a_1 + 2a_3 + 2^2a_5 + \ldots = 19$

By comparing $a_1 = 1, a_4 = 1, a_6 = 1$

and $a_2 = a_8 = a_{10} = 0$

$a_1 + 2a_3 + 2^2a_5 + 2^3a_7 + 2^4a_9 + \ldots = 19$

$a_1 = a_3 = a_9 = 1$

$a_5 = a_7 = a_{11} = a_{13} = 0$

$p(x) = 1 + x + x^3 + x^4 + x^6 + x^9$

$p(1) = 6$

67)

$(x - a)(x - b)(x - c) = d \begin{matrix} \nearrow \alpha \\ \rightarrow \beta \\ \searrow \gamma \end{matrix}$; $d \neq 0$

Now $(x - \alpha)(x - \beta)(x - \gamma) + d = 0$

$\Rightarrow (x - a)(x - b)(x - c) - d + d = 0$

$\Rightarrow (x - a)(x - b)(x - c) = 0$ roots are a, b, c

68)

Let α, β, γ by the roots of the given equation are in AP, that is

$\frac{1}{\alpha} + \frac{1}{\gamma} = \frac{2}{\beta}$

Adding 1/b to both the sides we get

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{3}{\beta}$

$\frac{\beta\gamma + \gamma\alpha + \alpha\beta}{\alpha\beta\gamma} = \frac{3}{\beta}$

$-\frac{54}{27} = \frac{3}{\beta}$

Since β is a root of the given equation we get

$10\beta^3 - a\beta^2 - 54\beta - 27 = 0$

substituting the value of β we get

$10\left(-\frac{3}{2}\right)^3 - a\left(-\frac{3}{2}\right)^2 - 54\left(-\frac{3}{2}\right) - 27 = 0$

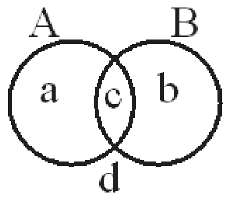
$10\left(-\frac{27}{8}\right) - \frac{9a}{4} + 81 - 27 = 0$

$\frac{9a}{4} = -\frac{10 \times 27}{8} + 54$

$$a = 4\left(-\frac{30}{8} + 6\right) = \frac{18}{2} = 9$$

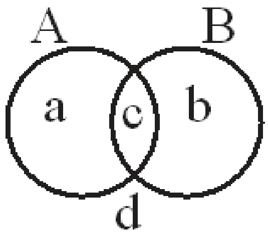
69)

S = {1, 2, 3, ...10}
P(S) = power set of S
$AR_1\ B \Rightarrow (A \cap \vec{B}) \cup (\vec{A} \cap B) = \phi$
R_1 is reflexive, symmetric
For transitive
$(A \cap \vec{B}) \cup (\vec{A} \cap B) = \phi$ {a} = ϕ = {b} A = B
$(B \cap \vec{C}) \cup (\vec{B} \cap C) = \phi$ □ B = C
□ A = C equivalence.



$$R_2 \equiv \left(A \cup \vec{B} \cup \vec{A} \cup B\right)$$

$R_2 \rightarrow$ Reflexive , symmetric
for transitive



$A \cup \vec{B} = \vec{A} \cup B \Rightarrow$ {a, c, d} = {b, c, d}
{a} = {b} □ A = B
$B \cup \vec{C} = \vec{B} \cup C \Rightarrow$ B = C
□ A = C
$\therefore\ A \cup \vec{C} = \vec{A} \cup C$
□ Equivalence

70)

$2^m - 2^n = 112$
$\Rightarrow 2^n(2^{m-n} - 1) = 16 \times 7$
$\Rightarrow 2^{m-n} = 8$
m – n = 3

71)

M = {5, 7, 9, 11, 12, 14, 16, 18, 20, 21, 23, 25, 27, 32}
S_2 – M = {2, 4, 6, 8, 10, 22, 24, 26, 28, 30}
Number of element = 10
Option (C).

72)

R_1 contains

$(2,2^0),(2,2^1),\ldots\ldots\ldots(2,2^6)$

$(3,3^0),(3,3^1)\ldots\ldots\ldots\ldots(3,3^4)$

$(5,5^0),(5,5^1)(5,5^2)$

$(7,7^0),(7,7^1)(7,7^2)$

$(11,11^0),(11,11^1)$

$(97,97^0),(97,97^1)$

R_2 contains $(2,2^0),(2,2^1),(2,2^2)$

$(3,3^0),(3,3^1),(3,3^2)$

$(5,5^0),(5,5^1),(5,5^2)$

.

.

$(97,97^0),(97,97^1)$

clearly R_2 is a subset of R_1; $R_1 \Delta R_2 = (R_1 - R_2) \cup (R_2 - R_1)$

$= R_1 - R_2$

$\Rightarrow \quad n(R_1 - R_2) = 6$

73)

$2(n(A) - n(AB)) = n(B) - n(AB)$
$2n(A) - n(B) = n(AB)$
$n(A) + 3n(B) = 5n(AB)$
$\Rightarrow n(A) + 3n(B) = 10n(A) - 5n(B)$

$8n(B) = 9n(A)$

$\frac{n(A)}{n(B)} = \frac{8}{9}$

$n(A) = 8k; n(B) = 9k$
$\Rightarrow n(AB) = 7k$
$n(A \cup B) = 10k \leqslant 10$

$k = 1$

74)

$n(E \cup H) = 60$

$n(E) = 40$

$n(H) = 30$

$n(E \cup H) = n(E) + n(H) - n(E \cap H)$

$60 = 40 + 30 - n(E \cap H)$

$n(E \cap H) = 10$

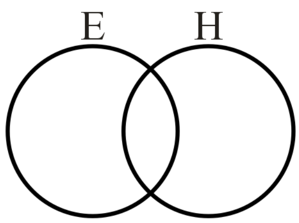
English and Hindi both $= \beta = 10$

only Hindi $= \alpha = n(H) - n(E \cap H) = 30 - 10 = 20$

x-int $= \alpha = 20$

y-int $= \beta = 10$

$\frac{x}{20} + \frac{y}{10} = 1$



$x + 2y = 20$

75)

Given series

$\left[\frac{a}{b}\right] + \left[\frac{2a}{b}\right] + \left[\frac{3a}{b}\right] + \ldots\ldots + \left[\frac{(b-1)\,a}{b}\right]$

have (b – 1) terms

First term + last term = $\left[\frac{a}{b}\right] + \left[\frac{ab}{b} - \frac{a}{b}\right]$

$= \left[\frac{a}{b}\right] + a + \left[-\frac{a}{b}\right]$

= a – 1

Similarly

second term + second last term

$= \left[\frac{2a}{b}\right] + \left[(b-2)\,\frac{a}{b}\right]$

$= \left[\frac{2a}{b}\right] + \left[a - \frac{2a}{b}\right]$

$= \left[\frac{2a}{b}\right] + a + \left[-\frac{2a}{b}\right]$

= a – 1

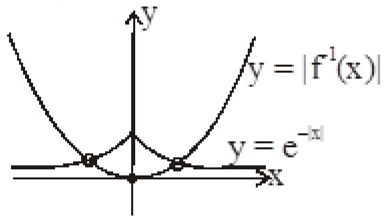
So required sum = $\frac{(b-1)}{2}\,(a-1)$

$\Rightarrow k_1 + k_2 = 3$

76) $f(x) = y = \ln(x + \sqrt{x^2+1})$

$\Rightarrow e^y = \sqrt{x^2+1} + x;\ e^{-y} = \sqrt{x^2+1} - x$

$f^{-1}(x) = \frac{e^x - e^{-x}}{2}$



No. of solutions = 2

77) (A) Even function

 (x – 2) and (x + 2) sgn(x + 2) both are even functions.

(B) Let $f(x) = \frac{1}{x\,(e^{x-1})} + \frac{1}{2x} \Rightarrow f(-x) = \frac{1}{-x\,(e^{-x}-1)} - \frac{1}{2x}$

 $f(-x) = \frac{e^x}{x\,(e^x-1)} - \frac{1}{2x}$.

 $f(x) - f(-x) = \frac{1-e^x}{x\,(e^x-1)} + \frac{1}{x} = -\frac{1}{x} + \frac{1}{x} = 0$

 $f(-x) = f(x) \Rightarrow$ even function.

(C) Let $g(x) = \log\left(\sin x + \sqrt{1+\sin^2 x}\right)$

$$g(-x) = \log\left(\sqrt{(1+\sin^2 x)} - \sin x\right)$$

$$= \log\left(\frac{1}{\sqrt{1+\sin^2 x} + \sin x}\right) = -g(x)$$

(D) clearly an even function.

78)

$$f(x) = \frac{\sqrt{x^2-25}}{(4-x^2)} + \log_{10}(x^2 + 2x - 15)$$

Domain : $x^2 - 25 \geq 0 \Rightarrow x \in (-\infty, -5] \cup [5, \infty)$

$4 - x^2 \neq 0 \Rightarrow x \neq \{-2, 2\}$

$x^2 + 2x - 15 > 0 \Rightarrow (x + 5)(x - 3) > 0$

$\Rightarrow x \in (-\infty, -5) \cup (3, \infty)$

$x \in (-\infty, -5) \cup (5, \infty)$

$\alpha = -5$; $\beta = 5$

$\alpha^2 + \beta^3$

79)

Here $f(x) = f(8-x) \Rightarrow f(x+4) = f(4 - x)$

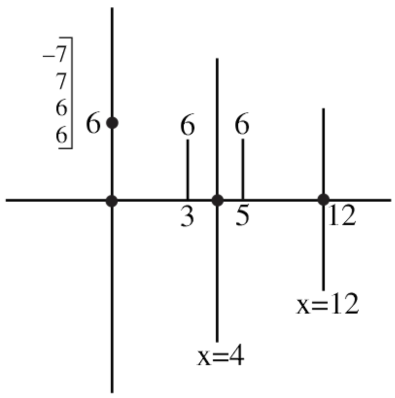
$f(x + 10) = f(14 - x) \Rightarrow f(x + 12) = f(12 - x)$

f(x) is symmetric about x = 4 and x = 12

so f(x) is periodic with period 16

f(0) = 6, f(3) = 6

f(8) = 6 f(5) = 6



Minimum number of α such that f(x) = 0

α can be 16 k, 16 k + 3, 16k + 5 or 16k + 8

minimum number of α = 7 + 7 + 6 + 6 = 26

80)

f(1) = 1 ...(1)

f(1 + 4) = f(1) + 16 + 32 = 49

f(5) = 49 ...(2)

$f(3 \times 5) = 9 + (5) + 8 = 9(49) + 8 = 441 + 8$

f(15) = 449

81)

$p(x) = x^2 + bx + c$

$p(p(1)) = p(p(2)) = 0$

$\Rightarrow p(1)$ & $p(2)$ are roots of $x^2 + bx + c = 0$

$-b = p(1) + p(2) \Rightarrow -b = 5 + 3b + 2c$ ...(1)

$c = p(1)p(2) \Rightarrow (1 + b + c)(4 + 2b + c) = c$ ...(2)

from (1) & (2)

get $b = \frac{-1}{2}$; $c = \frac{-3}{2}$

82)

$\alpha + \beta = 1154$ and $\alpha\beta = 1$

$(\sqrt{\alpha} + \sqrt{\beta})^2 = \alpha + \beta + 2\sqrt{\alpha\beta} = 1154 + 2 = 1154 = (34)^2$

$\Rightarrow \sqrt{\alpha} + \sqrt{\beta} = 34$

Again $(\alpha^{1/4} + \beta^{1/4})^2 = \sqrt{\alpha} + \sqrt{\beta} + 2(\alpha\beta)^{1/4} = 34 + 2 = 36$

$\alpha^{1/4} + \beta^{1/4} = 6$

83)

$$\frac{S_{14} + S_{11}}{S_{12} - 3S_{13}} = \frac{\alpha^{14} + \beta^{14} + \gamma^{14} + \alpha^{11} + \beta^{11} + \gamma^{11}}{\alpha^{12} + \beta^{12} + \gamma^{12} - \left[3\alpha^{13} + 3\beta^{13} + 3\gamma^{13}\right]}$$

$$= \frac{\left(\alpha^3 + 1\right) + \left(\beta^3 + 1\right) + \left(\gamma^3 + 1\right)}{\left(\alpha - 3\alpha^2\right) + \left(\beta - 3\beta^2\right) + \left(\gamma - 3\gamma^2\right)} = 2$$

use $\alpha^3 + 1 = 2\alpha - 6\alpha^2$

84)

$x^2 + (5 - a)x + a = 0$

$\alpha \quad \beta$

$\Rightarrow \quad \alpha + \beta = a - 5, \ \alpha\beta = a$

Now, $\alpha^3\beta + \alpha\beta^3 = 14(2 - a)$

$\Rightarrow \quad (a - 1)(a - 4)(a - 7) = 0$

∴ $a_1 = 1, a_2 = 4, a_3 = 7$

Now, $4x^2 + 4px + (4p - 3) > 0 \ \forall x \in R$

$\Rightarrow \quad (4p)^2 - 4(4)(4p - 3) < 0$

$\Rightarrow \quad p \in (1, 3)$

85) We have

$a^2 - (2x^2 + 1)a + x^4 + x = 0$

∴ $a = \frac{(2x^2 + 1) \pm \sqrt{(2x^2 + 1)^2 - 4(x^4 + x)}}{2}$

∴ $2a = (2x^2 + 1) \pm (2x - 1)$

positive sign $a = x^2 + x$

negative sign $2a = 2x^2 - 2x + 2$

$a = x^2 - x + 1$

if $x^2 + x - a = 0 \Rightarrow x = \frac{-1 \pm \sqrt{1+4a}}{2}$

if $x^2 - x + 1 - a = 0$

$\Rightarrow x = \frac{1 \pm \sqrt{1-4+4a}}{2} = \frac{1 \pm \sqrt{4a-3}}{2}$

for x to be real $a \geq 3/4$ and $a \geq -1/4$

$\Rightarrow a \geq 3/4 \Rightarrow 3 + 4 = 7$

86)

$1009 \square = \sum_{n=1}^{2020} [nx] = \sum_{n=1}^{2020} [(2021-n)x]$

$\sum_{n=1}^{2020} [1010 - nx]$

Also $[nx] + [1010 - nx] = 1010 + [nx] + [-nx]$

$= 1009$ ($nx \neq I$)

$(2 \times 1009)\square = \sum_{n=1}^{2020} 1009 \Rightarrow \square = 1010$

87)

$f(x+1) + f(x-1) = \sqrt{3}f(x), \forall x \in R$ .... (i)

Replacing x by x – 1 and x + 1 in (i), then

$f(x) + f(x-2) = \sqrt{3}f(x-1)$ .... (ii)

and $f(x+2) + f(x) = \sqrt{3}f(x+1)$ .... (iii)

Adding (ii) and (iii), we get

$2f(x) + f(x-2) + f(x+2) = \sqrt{3}(f(x-1) + f(x+1))$

$2f(x) + f(x-2) + f(x+2) = \sqrt{3}\left(\sqrt{3}f(x)\right)$

$f(x - 2) + f(x + 2) = f(x)$ .... (iv)

Replacing x by x + 2 in equation (iv)

$f(x) + f(x + 4) = f(x + 2)$ .... (v)

Adding equations (iv) and (v)

$f(x + 4) + f(x - 2) = 0$ .... (vi)

Again replacing x by x + 6 in equation (vi)

$f(x + 10) + f(x + 4) = 0$ .... (vii)

Substracting (vi) from (vii)

$f(x + 10) - f(x - 2) = 0$ .... (viii)

Replacing x by x + 2 in (viii)

$f(x + 12) - f(x) = 0$

$f(x + 12) = f(x)$

88)

$f : R \to R ; f(x) = \frac{x^3}{3} + (m-1)x^2 + (m+5)x + n$

$f'(x) = x^2 + 2(m-1)x + (m+5) \geqslant 0 \Rightarrow \Delta \leqslant 0$

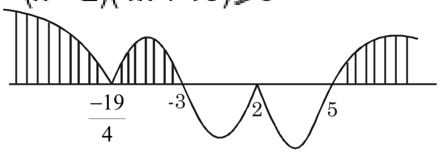
$\Rightarrow 4(m-1)^2 - 4(m+5) \leqslant 0$

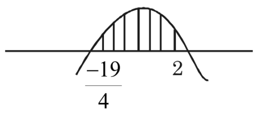
$\Rightarrow m^2 - 3m - 4 \leqslant 0$

$\Rightarrow\ (m-4)(m+1)\leqslant 0$
$\Rightarrow\ -1\leqslant m\leqslant 4$

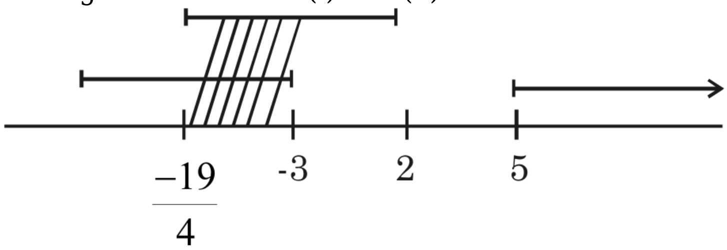
89)

$f(x)=\sqrt{\sqrt{38-4x^2-11x}.(x^2-2x-15)}$
for domain
$\sqrt{38-4x^2-11x}.(x^2-2x-15)\geqslant 0$ and $38-4x^2-11x\geqslant 0$
$\sqrt{-(x-2)(4x+19)}(x-5)(x+3)\geqslant 0 \quad -(4x^2+11x-38)\geqslant 0$
$-(x-2)(4x+19)\geqslant 0$





$x\in(-\infty,3]\cup\{2\}\cup[5,\infty)-(i)$ $\quad x\in\left[\frac{-19}{4},2\right]-(ii)$
Taking intersection of (i) and (ii)



Domain of f is $\left[\frac{-19}{4},-3\right]\cup\{2\}$
sum of integer in domain of f
s = (-4) + (-3) + 2 = -5
Thus |2 - (-5)| = |2+5| = 7

90)

$g(x)=x^{2015}+\operatorname{sgn}x+\left[\frac{x^2+1}{p}\right]$

for g(x) to be an odd function $\left[\frac{x^2+1}{p}\right]=0$

i.e. $0\leqslant\frac{x^2+1}{p}<1$ $\Rightarrow$ p is positive
$x^2+1<p$
$p>5\Rightarrow p=6$